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A weed is a plant growing where it is not wanted. Fat hen is a weed which grows in some crop fields.

Describe how you could estimate the **number** of fat hen plants in a field.

[5 marks]

[illegible]

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In potato plant fields, fat hen plants can grow up to a height of 2 m and absorb large quantities of nutrients from the soil.

Fat hen has a negative effect on the growth of potato plants.

Use the information provided to explain why.

In your answer, name the type of competition occurring between fat hen plants and potato plants.

[3 marks]

Question 5 continues on the next page

0 5 . 3 **Table 1** gives some features of fat hen seeds.

Table 1

Feature of seed	
Mean mass / mg	0.77
Maximum number produced per fat hen plant during a growing season	20 000
Percentage viable (able to develop) after a growing season	79

During a growing season, it was estimated that a total number of 550 fat hen plants grew on the fields of a farm.

Calculate the maximum mass, in kg, of viable fat hen seeds on this farm after a growing season.

Show your working.

Assume that all the seeds produced during the growing season remain on the farm.

[2 marks]

Answer _____ kg

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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